

Phishing Email Detection Program

With the rise of AI-generated phishing attacks, it is increasingly important to use modern AI tools, such as Hugging Face, to improve cyber deception detection and protect organizational networks.

Main Claim

AI-generated phishing attacks are becoming increasingly sophisticated and harder to detect as technology advances. These attacks can lead to stolen credentials, data breaches, and the loss of sensitive enterprise information. As these attacks evolve, it is becoming more difficult for employees to distinguish between legitimate business communications and malicious phishing attempts.

Key Method

To aid in phishing detection and protect organizational networks, we fine-tuned a Hugging Face model, bert-base-uncased, to identify potential phishing emails. The model was trained on a labeled dataset consisting of 100 real emails and 100 AI-generated phishing emails. This approach uses natural language processing (NLP) to analyze email content. The model's task is to classify emails as either safe or phishing.

Outcome



Email is received to an employee at ChirpyHub



Our AI model tokenizes the text and analyzes the language patterns



The model provides a prediction if the email is safe or needs to be reported for further analysis.

Key Findings

In this model, we achieved 100% accuracy on a dataset of AI-generated phishing emails and legitimate employee emails.

This indicates that the system can effectively distinguish between phishing and normal email communication.

However, it is important to note that this 90% accuracy is likely influenced by the relatively small size of the dataset used for testing.

Despite this limitation, deploying a system like this within an enterprise such as ChirpyHub could significantly reduce the risk of phishing attacks, protect clients, employees, and help automate aspects of cybersecurity defense.

Reccomendations

- Utilize and deploy modern AI tools as a defensive measure to improve cyber deception detection.
- Train and evaluate models on larger and more diverse datasets to improve performance.
- Improve systems to adapt to new AI-generated phishing tactics as they evolve.
- Enable employees to more easily report unusual or suspicious emails through detection systems.
- Allow security analysts to focus on more critical threats by automating routine phishing detection tasks.

Source Note

Below are sources used to complete this summary:

IBM. (2024). AI vs. human deceit: Unravelling the new age of phishing tactics. IBM Think. <https://www.ibm.com/think/x-force/ai-vs-human-deceit-unravelling-new-age-phishing-tactics>